

Design and Evaluation of an Adaptive Edge-Based Intrusion Detection System for Small IoT Networks



Network Guardian

//goodbye world

Members:

Ryan Hayden, Computer Engineering

Tobias Hill, Computer Engineering

Uriah Stover, Computer Science

Chloe Thacker, Computer Engineering

CS/EE 480 – Senior Capstone Project

09/08/2026

GitHub: <https://github.com/ucstove/NetworkGuardian>

Mentor:

Dr. Waseem Al Aqqad

Instructors:

Dr. Waseem Al Aqqad

Dr. Zahmeeth Sakkaff

1. ABSTRACT

The main goal of this project is to provide a low-cost intrusion detection/security system designed for a home/small business network that can be easily deployed from network to network. We will deploy a Machine Learning Model on an Internet of Things (IoT) LAN consisting of a Raspberry Pi and multiple ESP32 microcontroller units (MCUs). The Raspberry Pi will act as a packet sniffer in the LAN and will be used for the Machine Learning Model. The Raspberry Pi may also act as an access point on the network, providing a place where all of the embedded devices can connect to or have the Pi connect to the network, so that it will be easy for the consumer to use the product. This project will solve the problems of high-cost cybersecurity infrastructure and technical issues for smaller networks by implementing a more cost-effective solution using machine learning and low-cost MCUs.

2. PROBLEM STATEMENT

With the growing IoT networks in homes and small businesses, weak spots in LANs have increased. Often, IoT devices are not equipped with strong industrial cybersecurity measures, leaving important home appliances open to risk [1]. Artificial intelligence has enabled cyberattacks to occur more frequently than before. The rapid increase of attacks and vulnerabilities have left the regular homeowner and small business owner in need of an accessible and reliable cybersecurity system for their networks. Cybersecurity systems on LANs are often overpriced for the service provided. The main alternative is to make your own system, which requires technical skills in the field and therefore creates a barrier to entry in cybersecurity. This project is extremely relevant to the field of computer science and computer engineering due to the application of cybersecurity, machine learning, and embedded IoT MCUs.

3. BACKGROUND

Currently, there aren't many ways to protect home networks and small businesses without paying for expensive network security or installing firewalls on your network, which can be difficult for those who are technically limited. Physical firewalls are often expensive or have limited capabilities, like those made by Firewalla. The most affordable of these, costs \$289 and requires additional hardware and smartphone applications to operate [2]. Online software is often cloud-reliant and subscription-based, meaning it is not a one-time purchase. While AWS IoT Device Defender is a common IoT cybersecurity software, it runs entirely in the AWS cloud environment and requires users to pay a subscription fee each month [3]. Our solution would be an affordable system that can be purchased once and easily installed on a small network. This would allow the average person to be able to secure their network easily compared to other alternatives.

4. SYSTEM FEATURES

The system will include a machine learning model that will look at packets, traffic patterns, timing frequency, and behavior over time on the network. The models we are looking into are supervised multi-class models (Random Forest, XGBoost), one-class models (Isolation Forest, autoencoders), and semi-supervised models (Self-training, PU learning) . The model will be trained on a GPU workstation with the intention that training a model of this size will take significant resources and time to produce a quality model. The data we will use to train the model will come from the AWID2 dataset provided by Aegean University. This covers the majority of different attack classes which will help get a strong foundation to start with.

Additionally, we will be simulating our own attacks and normal internet traffic over the network we simulate. This is to help test if the model is able to adapt to different situations including those it hasn't encountered yet. This machine learning model will be imported onto a Raspberry Pi to read and detect the packets as they are sent throughout the network. The Raspberry Pi will also act as a server to store the data in an SQL database, which will allow for easy storage of relevant malicious data and information. This allows for the system to be disconnected from the internet and work independently. Attached to the Raspberry Pi will be an ALFA wifi antenna to pick up the network traffic from all devices, so that all packets sent can be scanned and checked. At this point, we are attempting to make it as easy as connecting the Raspberry Pi to the network, so that it will be easy for the consumer to use the product. The other direction we are exploring is setting up the Raspberry Pi as an access point on the network, providing a place where all of the embedded devices can connect to. MCUs with wifi chips will be used to model the devices on the IoT LAN. The ESP32s will be able to communicate with each other and send packets just like a network would. A web user interface will be implemented into the system as well. This UI will display any malicious data and information the machine learning model detects for the user to be aware of what's happening. It will also include the abnormalities, and the reason for why it is abnormal. The UI feature will be a locally hosted website, hosted on the Pi, that the user can directly connect to. The webpage will use javascript libraries such as React, and python backend framework Django to provide a high performance locally hosted dynamic webpage.

5. TIMELINE

This timeline starts from August 30th.

Week 1	Finalize the proposal and scope of project
--------	--

Week 2-6	Draft and Finalize Software Requirements Specifications (SRS)
Week 7-11	Draft and Finalize HLD
Week 12-13	Final Presentation & Report

6. BUDGET AND RESOURCES

Item	Cost	Link
ESP32 with Wifi Chips 5pk	\$24.39	Amazon.com
Raspberry Pi 5	\$259.95	Amazon.com
ALFA Network AWUS036ACM Wifi Antenna	\$59.99	Amazon.com
Breadboards	N/A	N/A
Total Cost	\$344.33	

7. REFERENCES

- [1] A. K. Ray and A. Bagwari, "IoT based Smart home: Security Aspects and security architecture," 2020 IEEE 9th International Conference on Communication Systems and Network Technologies (CSNT), Apr. 2020, doi: 10.1109/csnt48778.2020.9115737.
- [2] "Firewalla Purple SE: (500 Mbps) Cyber Security Firewall (Ships Worldwide)," *Firewalla*, 2017. <https://firewalla.com/collections/firewalla-products/products/firewalla-purple-se> (accessed Sept. 03, 2026).
- [3] "AWS IoT Device Defender Features - Amazon Web Services," *Amazon Web Services, Inc.*, 2024. <https://aws.amazon.com/iot-device-defender/features/> (accessed Sept. 03, 2026).